# Statistics in Health Sciences - Assignment 1

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# 1 Introduction

The aim of this assignment is to apply the framework proposed by Basagaña et al. to the data set copd in the file copd redux.RData. Data copd is a subset of 45 variables of the original data analyzed in the cited paper.

# 2 Analysis

## 2.1 Description of missingness

Provide the same missingness description than in the original analysis. Specifically, reproduce the following text that is shown in the original paper:

> "The subset used here included 342 participants and 85 variables. The range of missing data in the different variables went from 0% to 47.7%, with 68.2% of the variables having less than 5% of the data missing. Only 13.7% of the participants had complete data on all 85 variables. Overall, 5.9% of the information was missing; that is, 5.9% of the total of 29,070 cells ($85 \times 342 = 29,070$) had missing data."

Instructions:

- Note that, since you are working with a subset of the original data set, red numbers above are not expected to be the same than in your results.
- In your `Rnw` document, the text above must be able to automatically update all red numbers in the case of changing the values of `copd`. I.e., if data change, then all red numbers must also automatically update accordingly in your `pdf` when compiling your `Rnw`. As a hint, you can follow this procedure:
  1. For each red number, create an R object, which will depend on data and gives the desired quantity (i.e., if data change, the value in the object changes accordingly),
  2. To force rounding a quantity to a given decimal precision, you can use `sprintf()` (as seen in Exercise 0). Do not use `round()`,
  3. To get the thousands comma separator as in "29,070", you can use `prettyNum()`,
  4. Once you have an R object for each red number formatted as desired, use `\Sexpr{}` to insert each of them inline in LaTeX (as seen in Exercise 0).

```
> n_obs <- nrow(copd) # Number of participants
> n_vars <- ncol(copd) # Number of variables
> nans <- 100*colMeans(is.na(copd)) # Percentage of missing data by variable
> min_nan <- sprintf("%.2g", min(nans)) # MIRAR # Lower percentage
> max_nan <- sprintf("%.2g", max(nans)) # Upper percentage
>
> # Vairables with less than 5% of data missing (percentage)
> less_5 <- sprintf("%.2g", 100*length(nans[nans<5])/n_vars)
>
> # Percentage of complete cases
> comp_cases <- sprintf("%.2g", 100*mean(complete.cases(copd)))
>
> # Percentage of missing data
> miss_total <-  sprintf("%.2g", 100*mean(is.na(copd)))
>
> # Total cells observed (variables*observations)
> total_cells = prettyNum(n_obs*n_vars, big.mark = ",", decimal.mark = ".")
```

"The subset used here included 342 participants and 45 variables. The range of missing data in the different variables went from 0% to 32%, with 69% of the variables having less than 5% of the data missing. Only 33% of the participants had complete data on all 45 variables. Overall, 5.4% of the information was missing; that is, 5.4% of the total of 15,390 cells ($45 \times 342 = 15,390$) had missing data."

## 2.2 Multiple imputation

Apply multiple imputation to copd data using 100 imputations. Explain in a reasoned manner how you decided to set the values of the arguments `method` and `predictorMatrix` when using the `mice()` function. Show your code.

```
> # Display of default imputation methods for each variable
> ini <- mice(copd, maxit=0)
> ini

## Class: mids
## Number of multiple imputations:  5
## Imputation methods:
##                id  index_barthel2v1    quo_calcpostv1         rv_prepv1        incp_fevv1
##                ""             "pmm"                ""             "pmm"             "pmm"
##       fevp_postv1          rv_tlcv1        ctnf_s_mv1    vtdvdbid_ecov1          pp_ecov1
##                ""             "pmm"             "pmm"             "pmm"             "pmm"
##          sin11_v1           vas2_v1         ic_prepv1          iam_chv1          sin15_v1
##          "logreg"             "pmm"             "pmm"          "logreg"          "logreg"
##        cil8_s_mv1           sin7_v1           sin8_v1             ch_v1         vc_prepv1
##             "pmm"          "logreg"          "logreg"             "pmm"             "pmm"
##       dtdve_ecov1          cccs1_v0      kcals_set_v0         impactsv1          monoA_v1
##             "pmm"          "logreg"             "pmm"             "pmm"             "pmm"
##  lvmass_ase_ecov1        vtit_catv1         urg_ambv0            po2_v1           ahpf_v1
##             "pmm"          "logreg"                ""             "pmm"             "pmm"
##            pro_v1           basA_v1           fc_pmv1            tad_v1            col_v1
##             "pmm"             "pmm"             "pmm"             "pmm"             "pmm"
##           sin5_v1             bc_v1       fvcp_postv1           vas1_v1           sin6_v1
##          "logreg"             "pmm"                ""             "pmm"          "logreg"
##       graudisn_v1 cpcr_mg_l_mediav1          neutA_v1           ahpl_v1        activityv1
##         "polyreg"             "pmm"             "pmm"             "pmm"             "pmm"
## PredictorMatrix:
##                  id index_barthel2v1 quo_calcpostv1 rv_prepv1 incp_fevv1 fevp_postv1
## id                0                1              1         1          1           1
## index_barthel2v1  1                0              1         1          1           1
## quo_calcpostv1    1                1              0         1          1           1
## rv_prepv1         1                1              1         0          1           1
## incp_fevv1        1                1              1         1          0           1
## fevp_postv1       1                1              1         1          1           0
##                  rv_tlcv1 ctnf_s_mv1 vtdvdbid_ecov1 pp_ecov1 sin11_v1 vas2_v1 ic_prepv1
## id                      1          1              1        1        1       1         1
## index_barthel2v1        1          1              1        1        1       1         1
## quo_calcpostv1          1          1              1        1        1       1         1
## rv_prepv1               1          1              1        1        1       1         1
## incp_fevv1              1          1              1        1        1       1         1
## fevp_postv1             1          1              1        1        1       1         1
##                  iam_chv1 sin15_v1 cil8_s_mv1 sin7_v1 sin8_v1 ch_v1 vc_prepv1 dtdve_ecov1
## id                      1        1          1       1       1     1         1           1
## index_barthel2v1        1        1          1       1       1     1         1           1
## quo_calcpostv1          1        1          1       1       1     1         1           1
## rv_prepv1               1        1          1       1       1     1         1           1
## incp_fevv1              1        1          1       1       1     1         1           1
## fevp_postv1             1        1          1       1       1     1         1           1
##                  cccs1_v0 kcals_set_v0 impactsv1 monoA_v1 lvmass_ase_ecov1 vtit_catv1
## id                      1            1         1        1                1          1
## index_barthel2v1        1            1         1        1                1          1
## quo_calcpostv1          1            1         1        1                1          1
## rv_prepv1               1            1         1        1                1          1
## incp_fevv1              1            1         1        1                1          1
## fevp_postv1             1            1         1        1                1          1
##                  urg_ambv0 po2_v1 ahpf_v1 pro_v1 basA_v1 fc_pmv1 tad_v1 col_v1 sin5_v1
## id                       1      1       1      1       1       1      1      1       1
## index_barthel2v1         1      1       1      1       1       1      1      1       1
## quo_calcpostv1           1      1       1      1       1       1      1      1       1
## rv_prepv1                1      1       1      1       1       1      1      1       1
## incp_fevv1               1      1       1      1       1       1      1      1       1
## fevp_postv1              1      1       1      1       1       1      1      1       1
##                  bc_v1 fvcp_postv1 vas1_v1 sin6_v1 graudisn_v1 cpcr_mg_l_mediav1 neutA_v1
## id                   1           1       1       1           1                1        1
## index_barthel2v1     1           1       1       1           1                1        1
```

```
## quo_calcpostv1       1           1       1       1           1                 1        1
## rv_prepv1            1           1       1       1           1                 1        1
## incp_fevv1           1           1       1       1           1                 1        1
## fevp_postv1          1           1       1       1           1                 1        1
##                  ahpl_v1 activityv1
## id                     1          1
## index_barthel2v1       1          1
## quo_calcpostv1         1          1
## rv_prepv1              1          1
## incp_fevv1             1          1
## fevp_postv1            1          1
```

We can see that the mice package suggest imputing initialy with de mean for continuous variables and with a logistic regression for binary ones. Further research on the dataset has shown that the variables have no deterministic relation. So we do not change the imputation methods. We have a variable (`id`) that don't have any role in the imputation process, so we change `predictorMatrix` and `methods` acording to that.

```
> # Change in parameters
> meth <- ini$method
> pred <- ini$predictorMatrix
>
> meth["id"] <- "" # do not impute "id"
> pred["id", ] <- 0 # we don't want "id" to be imputed
> pred[, "id"] <- 0 # we don't want "id" to be predictor
```

```
> # Imputation of 100 datasets
> M <- 100 # Number of imputations for each missing value
> imp <- mice(data = copd,
+             m = M,
+             method = meth,
+             predictorMatrix = pred,
+             maxit = 20,
+             printFlag = FALSE,
+             seed = 666)
> # imp£imp (output llarg)
> #summary(imp)
```

## 2.3 Integration of multiple imputation in cluster analysis

Reproduce the original analysis in the paper using the miclust package to integrate multiple imputation in the cluster analysis. Apply the proposed framework using the same setting than in the original analysis. To do that, follow these steps:

1. Read the help of the `getdata()` function in the `miclust` package. Then, accordingly, create a *list* of *data.frames* and pass it to the `getdata()` function. Give a detailed explanation of the data processing you did to get that *list*. Show your code.

```
> # ?miclust::getdata
> # getdata() requires the data to be a list or data.frame where the first element is the raw data:
>
>   #1- prepare raw data (all numeric and without id)
> copd_raw <- copd
> copd_raw$id <- NULL
> num_cols <- sapply(copd_raw, function(x) is.numeric(x) || is.integer(x) || is.logical(x))
> copd_raw <- copd_raw[, num_cols]
>
>   #2- prepare imputated data
> imp_list <- complete(imp, action = "all") # extracts the completed data from a mids object
> imp_list_clean <- lapply(imp_list, function(df) {
+   df$id <- NULL
+   df <- df[, num_cols]
+   return(df)
+ })
```

```
>
>   #3- create list where the first element is the raw data
> my_cluster_list <- c(list(copd_raw), imp_list_clean)
>
> # prepare the data for miclust
> imp_midata <- miclust::getdata(data = my_cluster_list)
>
> # now we check
> class(imp_midata)

## [1] "list"   "midata"

> length(imp_midata$impdata)

## [1] 100
```

2. Use the `miclust()` function to perform the integration of missing data in cluster analysis using the same settings as in the original analysis. Show your code and explain how and why you set all arguments in the `miclust()` function.

```
> miclust_1 <- miclust(data = imp_midata,
+                      method = "kmeans",
+                      search = "backward",
+                      ks = 2:5,
+                      usedimp = 1:100,
+                      seed = 666)

## . . . . imp   5 . . . . imp  10 . . . . imp  15 . . . . imp  20
## . . . . imp  25 . . . . imp  30 . . . . imp  35 . . . . imp  40
## . . . . imp  45 . . . . imp  50 . . . . imp  55 . . . . imp  60
## . . . . imp  65 . . . . imp  70 . . . . imp  75 . . . . imp  80
## . . . . imp  85 . . . . imp  90 . . . . imp  95 . . . . imp 100
##
##  Analysis done.

> #summary(miclust_1)
> miclust_1$kfin # optimal k (number of clusters)

## [1] 2
```

- Comentar perque no utilitzem l'id per fer kmeans
- Comentar com preparem les dades per pasarles a getdata() (info executant ?mice::getdata())
- Explicar perque fem search backward (està a l'article)
- Trobar explicació de kmax=5 (vaig copiar la simulació de l'article)

# 3 Results

Using `plot()` and `summary()` methods in the `miclust` package, provide plots and tables for the results similar than those shown in the original paper. Also, show a barplot for the percentage of times each variable is selected among the imputed data sets, in decreasing order. Explain how you set all arguments in all functions and methods you use. Show your code.

```
> # Distribution of the CritCF criterion over the imputed data sets, one box per
> # explored number of clusters (ks = 2:5);
> plot(miclust_1, metric = "critcf", col.critcf = "darkmagenta")
```

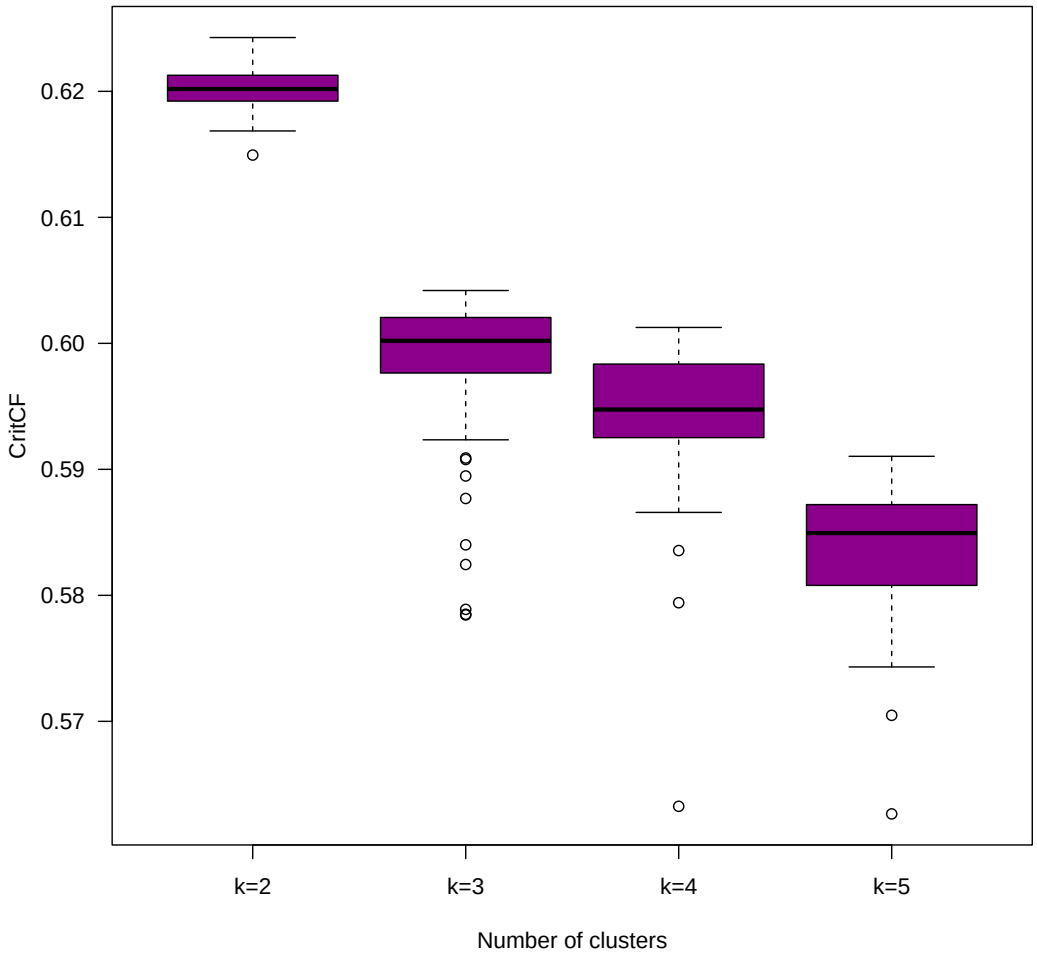



**Figure 1:** Boxplot of CritCF by number of clusters ($k$) over the 100 imputed data sets. Each box is based on 100 values; higher CritCF is preferred.

```
> # Final number of clusters selected by miclust
> kfin <- miclust_1$kfin
>
> # Percentage of appearance of each variable (metric = "varsel") for k = kfin
> plot(miclust_1, k = kfin, metric = "varsel", col.varsel = "skyblue")
```

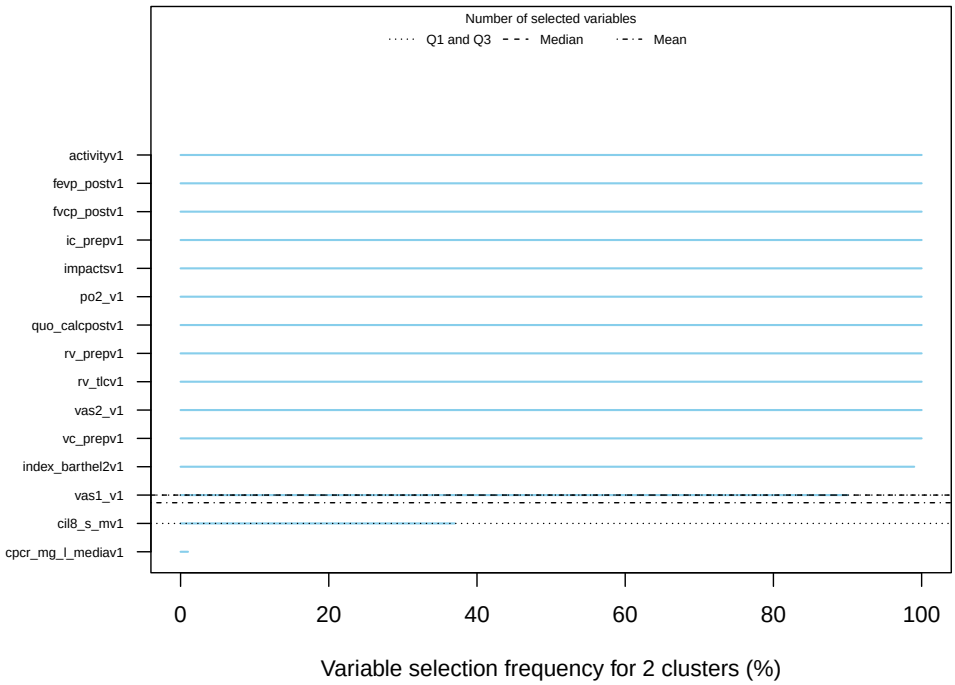



**Figure 2:** Variables that remained in the final set of variables in at least one of the 100 imputed data sets after the backward selection algorithm for $k = 2$, by percentage of appearance. Horizontal lines show the first and third quartiles, median and mean of the number of selected variables.

```
> # Set quantilevars = 0.5 so the numbers of final variables equals the median
> # number of variables selected across the imputed data sets.
> summary_k <- summary(miclust_1, quantilevars = 0.5)
>
> summary_kp1 <- summary(miclust_1, k = kfin + 1, quantilevars = 0.5)
```

```
> # Distribution of the probability of assignment to each cluster, for k = kfin
> a1 <- as.matrix(summary_k$allocationprobabilities)
>
> # Same distribution, for k = kfin + 1
> a2 <- as.matrix(summary_kp1$allocationprobabilities)
>
> # Numeric part, an empty row before each k block, followed by the rows of
> # the corresponding summary
> vals <- rbind(rep(NA, 5), a1, rep(NA, 5), a2)
>
> # Text part, one label per row.
> labels <- c(paste0("$k=", kfin, "$"),
+             paste0("Cluster ", seq_len(nrow(a1))),
+             paste0("$k=", kfin + 1, "$"),
+             paste0("Cluster ", seq_len(nrow(a2))))
>
> # Combine text part, first column, and numeric values into one data.frame
> tab_alloc <- data.frame(labels, vals)
>
```

```
>
> # Create the xtable object. 'digits' has 7 elements: the 1st is for the
> #automatic row names (not printed), the 2nd for the text labels (no decimals
> # apply), and the last 5 for the numeric columns (2 decimals each).
> # The caption and the label are defined here.
> alloc <- xtable(tab_alloc,
+                 digits = c(0, 0, 2, 2, 2, 2, 2),
+                 align = c("l", "l", "c", "c", "c", "c", "c"),
+                 caption = paste0("Distribution of the frequency of assignment
                         (proportion of the ", length(miclust_1$usedimp),
+                          " imputed data sets) to each cluster, for $k=",
+                          kfin, "$ and $k=", kfin + 1, "$."),
+                 label = "tab:alloc")
>
> # Column names, centered and in bold, the first one is left blank
> colnames(alloc) <- c("","\\multicolumn{1}{c}{\\bf Minimum}",
+                          "\\multicolumn{1}{c}{\\bf First quartile}",
+                          "\\multicolumn{1}{c}{\\bf Median}",
+                          "\\multicolumn{1}{c}{\\bf Third quartile}",
+                          "\\multicolumn{1}{c}{\\bf Maximum}")
>
> # Print the table un LaTeX
> print(alloc,
+       table.placement = "ht",
+       size = "footnotesize",
+       include.rownames = FALSE, # Row names are not printed, labels are in column 1
+       include.colnames = TRUE,
+       floating = TRUE,
+       sanitize.text.function = force,
+       booktabs = TRUE)
```

| | Minimum | First quartile | Median | Third quartile | Maximum |
|---|---|---|---|---|---|
| $k = 2$ | | | | | |
| Cluster 1 | 0.50 | 1.00 | 1.00 | 1.00 | 1.00 |
| Cluster 2 | 0.57 | 1.00 | 1.00 | 1.00 | 1.00 |
| $k = 3$ | | | | | |
| Cluster 1 | 0.64 | 1.00 | 1.00 | 1.00 | 1.00 |
| Cluster 2 | 0.56 | 1.00 | 1.00 | 1.00 | 1.00 |
| Cluster 3 | 0.78 | 1.00 | 1.00 | 1.00 | 1.00 |

**Table 1:** Distribution of the frequency of assignment (proportion of the 100 imputed data sets) to each cluster, for $k = 2$ and $k = 3$.

```
> # Frequency of selection of each variable across the imputed data sets
> y   <- getvariablesfrequency(miclust_1)
>
> # Sort the variables by decreasing percentage of selection
> y_ord <- order(y$percfreq, decreasing = TRUE)
> perc <- y$percfreq[y_ord]
> name <- y$varnames[y_ord]
>
> # Bars drawn as thick vertical lines (type = "h", lwd = 20).
> plot(perc, type = "h", main = "", xlab = "", ylab = "Percentage of times selected",
+      xlim = 0.5 + c(0, length(perc)), ylim = c(0, 100),
+      lwd = 20, col = "skyblue", xaxt = "n")
>
> # Variable names below each bar, las = 2 draws them vertically, and
> # cex.axis = 0.9 shrinks them slightly so that the long names fit
> axis(1, at = 1:length(name), labels = name, las = 2, cex.axis = 0.9)
```

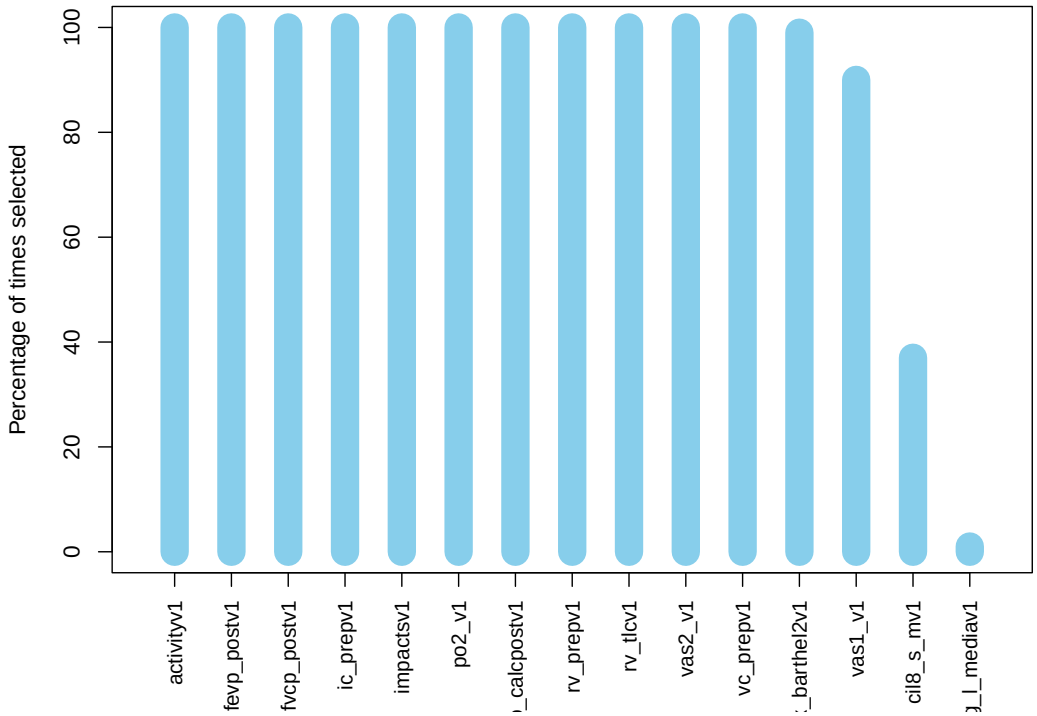



**Figure 3:** Percentage of the 100 imputed data sets in which each variable was selected by the backward algorithm for $k = 2$, in decreasing order.

```
> # This code is adapted from the miclust README, which can be accessed at
> # https://cran.r-project.org/web/packages/miclust/readme/README.html
```